KOLHAPUR INSTITUTE OF TECHNOLOGY'S
COLLEGE OF ENGINEERING KOLHAPUR
(EMPOWERED AUTONOMOUS)

COMPUTER SOCIETY OF INDIA
ESTD. 1965
सर्वे भवन्तु सुखिनः

PROJECT EXPO 2026

Smart Traffic Management System

Dynamic Traffic Signal Control

Team Name : FlowForce

Department: Computer Science and Engineering | Year: 3rd Year

Guide: Prof. Yogita Narule

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Problem Statement

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The Problem

Traffic signals often operate on predetermined timings that may not reflect the actual number of vehicles waiting in each lane.

1. **Long waiting times -** Vehicles wait even when traffic demand differs between lanes.
2. **Uneven congestion -** One approach may have a long queue while another has fewer vehicles.
3. **Emergency delays -** Congestion can delay emergency vehicles travelling through intersections.

Existing Solutions & Gaps

Existing approach	Limitation
Fixed-time signals	Do not adapt their timing to changing traffic demand.
Manual traffic control	Requires continuous human monitoring and intervention.
Emergency vehicle priority systems	May require dedicated communication and compatible infrastructur
CCTV-based traffic monitoring	Monitoring alone does not automatically optimize signal timings

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Project Idea & Objectives

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Our Idea

A SUMO-based traffic system that dynamically adjusts signal timings and prioritizes emergency vehicles using Python and TraCI.

What is unique/innovative?

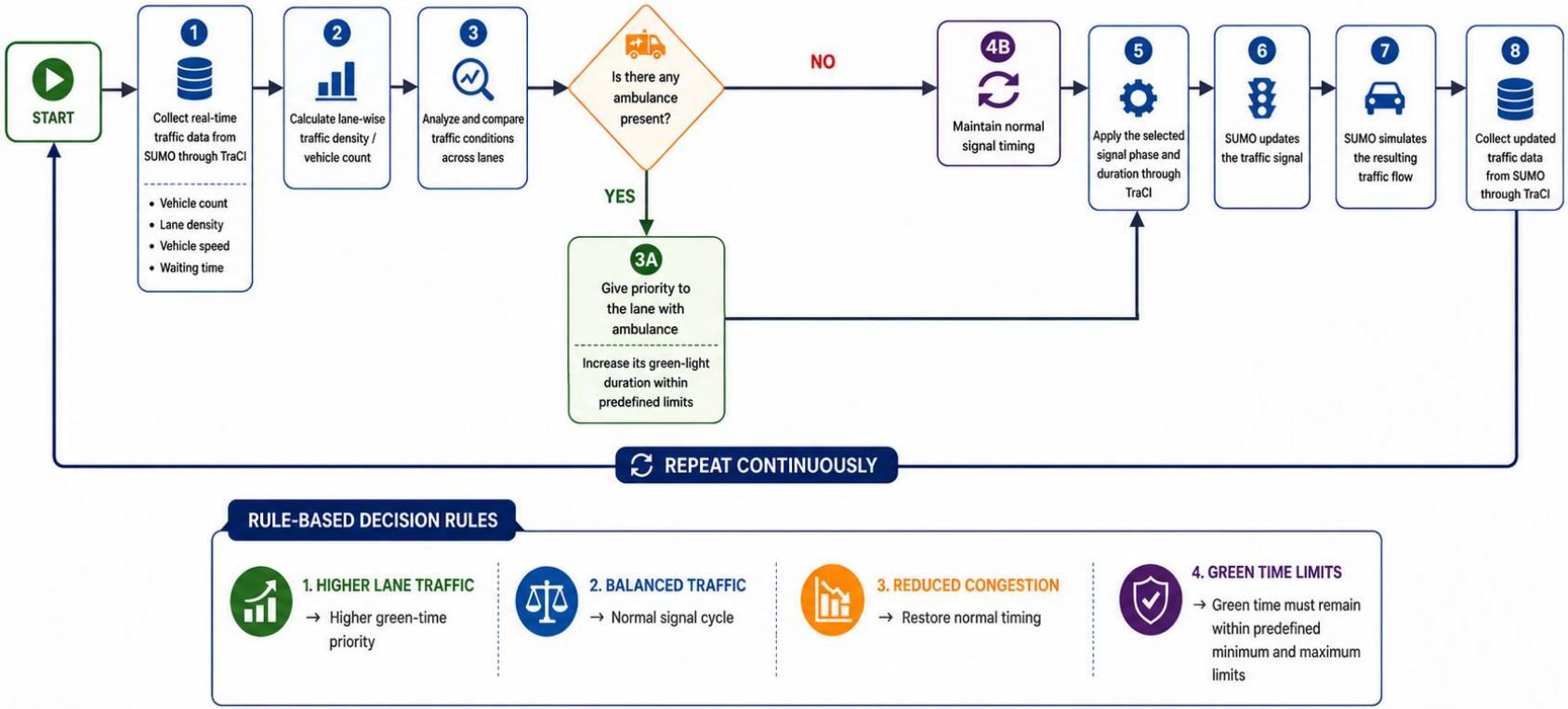
1. **Traffic-responsive signals** - Green-light duration changes based on simulated traffic conditions rather than relying only on fixed timings.
2. **Emergency-vehicle priority** - Includes a rule-based priority mechanism for emergency vehicles in the simulation
3. **Testable simulation environment** - SUMO and TraCI allow traffic scenarios and signal-control decisions to be tested in a simulated road network.

Objectives

1. Adapt signal timings to simulated vehicle counts, lane density, speed, and waiting time.
2. Reduce unnecessary vehicle waiting through demand-based signal decisions.
3. Implement emergency-vehicle priority using predefined rules.
4. Evaluate traffic-control behaviour through simulation scenarios.

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Prototype & Working



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Technology Stack & Impact

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Tools & Technologies

Simulation:





Programming:





Control logic: Rule-based decision engine

Impact & Future Scope

Potential Impact

- Better allocation of green time
- Reduced unnecessary waiting
- Emergency-vehicle priority

Future Scope

- Test on more complex road networks
- Compare against fixed-time signals
- Multi-intersection coordination
- Explore real-world integration and deployment

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Results & Conclusion

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Fixed-time Control

126 sec

Average waiting time

Dynamic-time Control

79 sec

Average waiting time

Optimization

37.3%

Reduction in waiting time

Conclusion

The Smart Traffic Management System uses SUMO, TraCI, and Python to dynamically adjust traffic signal timings based on vehicle count, vehicle type, emergency vehicle and waiting time. The rule-based approach enables traffic-responsive signal control and prioritizes emergency vehicles within the simulated road network.

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Thank You

Questions & Demo